

Time Series, Part 1

Practical worksheet

CS3621 Data Mining · to be done after Lecture 1 · about 2 hours · questions only

HOW TO USE THIS

Four parts, and they get harder. **Part A** is pure plot reading and needs no computer at all: print it, or work from the screen with a sheet of paper. **Part B** is short written answers. **Part C** is at the keyboard, using the dataset supplied with this worksheet. **Part D** is a critique exercise, and it is the one that will show up in the exam in some form.

Answers are not included in this version. Model solutions will be released after the submission deadline. Work every question before you look at anything else: getting an answer wrong and then finding out why is worth far more than reading a correct answer first.

Marks in the margin are indicative weights, not real marks. Total 100.

PART	WHAT IT TRAINS	NEEDS A COMPUTER?	TIME	WEIGHT
A. Read the plot	Recognising components, memory, stationarity and periodicity by eye	No	35 min	32
B. Reason it out	Explaining <i>why</i> , in your own words	No	30 min	28
C. Hands on	The full pipeline on real-shaped data	Yes	45 min	28
D. Find the mistakes	Reviewing somebody else's analysis	Optional	20 min	12

WHAT YOU NEED FOR PART C, AND WHAT YOU DO NOT

Required: Python with `numpy` `pandas` `matplotlib` `scipy` `statsmodels`. That is all.

The dataset. `clinic_attendances.csv` is provided alongside this sheet. **If you cannot get hold of it for any reason, you are still not blocked:** the box at the start of Part C contains a short cell that generates the identical file locally, with no download. Use whichever is easier.

Optional, and not required to complete anything here:

`CS3621_TimeSeries_Companion.ipynb`, the lecture companion notebook. It contains worked code for every technique from Lecture 1, already run with its outputs saved. It is a reference and a safety net, not a dependency. Parts A, B and D need no computer at all.

PART A · READ THE PLOT

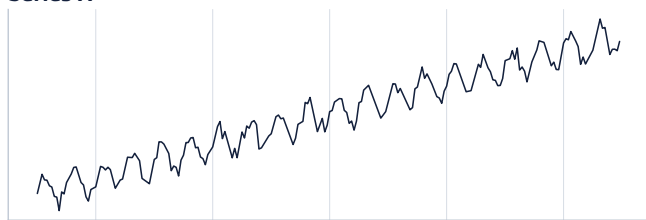
NO COMPUTER · 35 MIN · 32 MARKS

A1. What is each series made of?

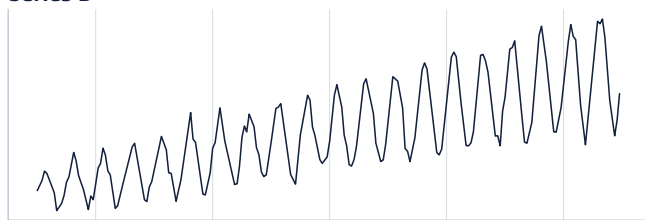
8 marks

Four series, each 20 years of monthly data.

Series A



Series B



Series C



Series D



For **each** of A, B, C and D, write down:

1. which components are present: **trend**, **seasonality**, **cycle**, or none of them;
2. if there is seasonality, is it **additive** or **multiplicative**, and how can you tell from the picture alone;
3. one sentence on what kind of real quantity might behave that way.

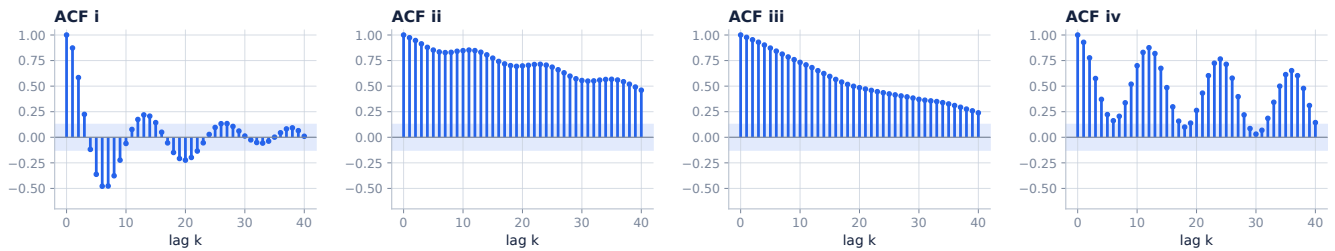
HINT

For part 2, put a ruler along the tops of the peaks and another along the bottoms of the troughs. Parallel or fanning?

A2. Match each series to its ACF

8 marks

Here are the four autocorrelation functions of those same four series, in a scrambled order.



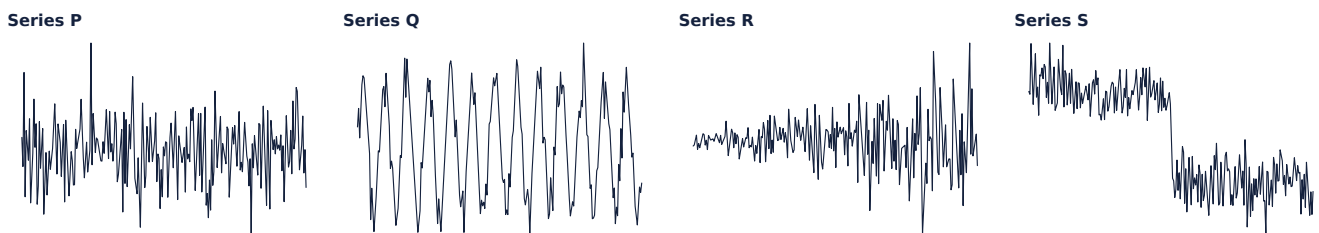
Match **i, ii, iii, iv** to **A, B, C, D**. For each match, give the **one visual feature** that settles it. Answers of the form "it just looks like it" score nothing.

HINT

Ask three questions of each ACF, in this order. Does it decay slowly and almost straight, or quickly? Does it ever go clearly **negative**? Are there **bumps at regular intervals**, and if so, what is the spacing?

A3. Which are stationary?

8 marks



For each of P, Q, R and S: stationary, yes or no? If no, state **which of the three conditions fails**, and what you would do about it.

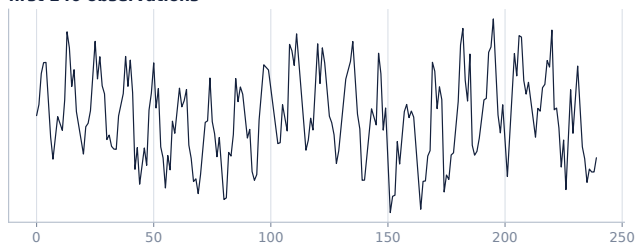
Then the harder half of the question. Below are the p-values from the two tests. For **two** of these four the tests agree with your eyes, and for **two** they do not. **Explain each disagreement.**

SERIES	ADF P	KPSS P	WHAT THE TWO TESTS TOGETHER CONCLUDE
P	< 0.001	0.10	stationary
Q	< 0.001	0.10	stationary
R	< 0.001	0.10	stationary
S	0.862	0.01	not stationary

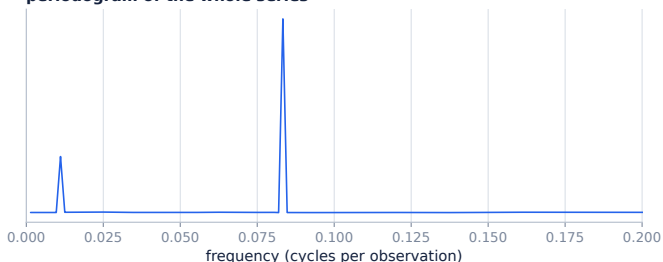
A4. Read the periodogram

4 marks

first 240 observations



periodogram of the whole series



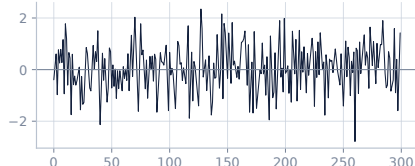
1. How many periodic components are in this series, and what is the **period** of each? Show the arithmetic.
2. Only one of them is visible to the eye in the left-hand panel. Which, and why is the other one so hard to see there?
3. If these were monthly observations, would you call either of these components **seasonality**? Justify.

A5. Judge the remainder

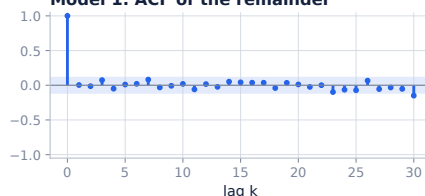
4 marks

Two people decompose the same series and show you the remainder from their model.

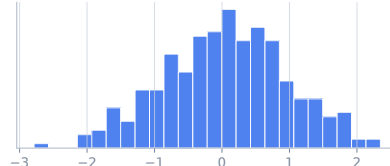
Model 1: remainder over time



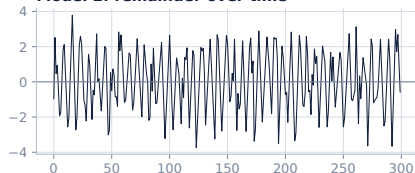
Model 1: ACF of the remainder



Model 1: distribution



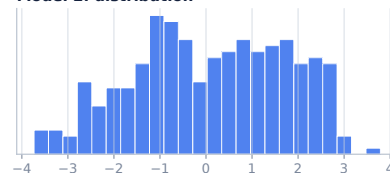
Model 2: remainder over time



Model 2: ACF of the remainder



Model 2: distribution



Which model is finished, and which is not? Name **every** piece of evidence you can see, and for the unfinished one say specifically what structure has been left behind and roughly what its period is.

B1. The shuffle test, in your own words

4 marks

In the lecture we shuffled the airline series and the histogram did not change at all. Explain, in no more than four sentences, **why no statistic computed from the histogram can ever detect the difference**, and name one statistic that **can**.

B2. Season or cycle?

6 marks

Classify each of the following as **seasonality**, a **cycle**, or **neither**, and give the one-line test you applied.

1. Ice cream sales, monthly, in Colombo.
2. The number of sunspots, annually, with peaks roughly 11 years apart.
3. Electricity demand, hourly, over one month.
4. The price of rice after a drought year.
5. Hospital admissions on the day after a public holiday.
6. The boom and bust of the semiconductor industry, roughly every 4 to 6 years.

B3. Why did differencing not finish the job?

5 marks

On the airline data we took logs and then one difference, and the ADF p-value was still 0.071. Explain what was still in the series, why a **first** difference cannot remove it, and what does remove it. Be precise about which lag is involved and why.

B4. The test that was wrong

4 marks

You are given a series whose mean is dead flat but whose swings get visibly wider every year. ADF returns $p < 0.001$. Your project partner writes "ADF confirms the series is stationary, so we proceeded to fit ARIMA".

Write the two-sentence correction you would put in the review comments.

B5. Order of operations

4 marks

Why do we stabilise the **variance** before we remove the **trend**, and not the other way round? Give a concrete reason, not "because the slides said so".

B6. The anomaly a threshold cannot find

5 marks

A clinic records 165 attendances on a Sunday. The lowest weekday count all year is 168 and the highest is 260, so 165 is not an extreme value in any global sense. Yet it is a serious anomaly.

1. Which of the three kinds of anomaly is this?
2. Explain why **no** threshold applied to the raw series can catch it, however cleverly you choose the threshold.
3. Describe, in two steps, a procedure that does catch it.

THE DATA

`clinic_attendances.csv`: daily attendance counts at an outpatient clinic, 3 January 2022 to 1 January 2025, 1,095 rows, two columns, `date` and `attendances`. It is **synthetic**, which is deliberate: I know exactly what is in it, so the solutions can tell you whether you found everything.

```
# start here
import numpy as np, pandas as pd, matplotlib.pyplot as plt
s = pd.read_csv('clinic_attendances.csv', parse_dates=[0], index_col=0).squeeze()
s.index.freq = 'D'
s.plot(figsize=(12,3))
```

IF YOU CANNOT GET THE CSV, YOU ARE NOT BLOCKED

Run the cell below instead. It **generates the identical dataset**, byte for byte, with no download and no internet. Everything after this point works exactly the same way, so you can start Part C right now.

```
import numpy as np, pandas as pd, matplotlib.pyplot as plt

rng = np.random.default_rng(404)
n = 1095
t = np.arange(n)
idx = pd.date_range("2022-01-03", periods=n, freq="D")

trend = 120 + 55 * (1 - np.exp(-t / 420))
weekly = np.array([1.18, 1.10, 1.06, 1.04, 0.98, 0.42, 0.34])[idx.dayofweek]
yearly = 1 + 0.16 * np.sin(2 * np.pi * (t - 40) / 365.25)
y = trend * weekly * yearly * rng.normal(1, .055, n)

# four deliberate faults, do not read this too closely before you attempt C4
y[250] *= 1.85
y[600:612] = trend[600:612] * .995
sun = np.where(idx.dayofweek == 6)[0]; j = int(sun[sun > 800][3]); y[j] = trend[j]
y[930:945] *= np.linspace(1.0, 1.45, 15)

s = pd.Series(np.round(y).astype(int), index=idx, name="attendances")
s.index.name = "date"; s.index.freq = "D"
```

```
s.to_csv("clinic_attendances.csv") # now you have the file too
print(len(s), "rows,", s.index[0].date(), "to", s.index[-1].date())
```

C1. Look first

5 marks

1. Plot the whole series, then plot **only the first eight weeks**. What do you see in the second plot that is invisible in the first?
2. Write two sentences naming every component you can identify, with the period of each.
3. Is the seasonality additive or multiplicative here? Give the evidence.

C2. Find the periods without assuming them

6 marks

Compute the periodogram. Remember the caveat from the lecture about what you must do to the series first.

```
y = s.values.astype(float)
y = y - y.mean() # and what else? see the lecture
power = np.abs(np.fft.rfft(y))**2 / len(y)
freq = np.fft.rfftfreq(len(y), 1.0)
# plot power against freq, then convert the peaks to periods
```

1. Which periods does it find? Report them in days, and say what each one means in plain English.
2. What happens to the plot if you **skip** the de-trending step? Try it and describe the difference in one sentence.

C3. Decompose

6 marks

```
from statsmodels.tsa.seasonal import STL
res = STL(s, period=7, robust=True, seasonal=15, trend=181).fit()
res.plot()
```

1. Run it. Describe each of the four panels in one sentence.
2. Run it again with `robust=False`. Where do the two **trend** estimates differ most, and why there?
3. This series has **two** seasonal periods. STL removed one of them. Where did the other one go? Point at the panel it ended up in.

C4. Hunt the anomalies

6 marks

There are **four** deliberate faults in this dataset. Find them.

```
rem = s - res.seasonal - res.trend
z = (rem - rem.median()) / (1.4826 * np.median(np.abs(rem - rem.median())))
flagged = s.index[np.abs(z) > 5]
```

1. List the dates you flag at a threshold of 5, and group them into distinct incidents.
2. For each incident, say which of the three kinds of anomaly it is.
3. Repeat at thresholds 3, 4 and 6. Tabulate how many flags are real and how many are false alarms at each. Which threshold would you ship, and what does your choice assume about the cost of a false alarm?
4. Now recompute `z` using `rem.mean()` and `rem.std()` instead of the median and MAD. What changes, and what is that failure called?

C5. Make it stationary

5 marks

Build the treatment pipeline and report the evidence at every stage.

```
from statsmodels.tsa.stattools import adfuller, kpss
# H0(ADF) = unit root, want SMALL p      |      H0(KPSS) = stationary, want LARGE p
```

1. Produce a table with one row per stage, reporting **ADF p** and **KPSS p**: raw, after a log, after a first difference, after a seasonal difference at lag 7. At which stage does the series first pass **both** tests?
2. Plot the ACF of the series at the stage where it first passes. Look at **lag 7**. Do you still believe the tests?
3. Now try a **second route**: log, then go straight to the seasonal difference at lag 7, with **no** first difference at all. Does it pass?
4. **Compare the two routes** using the two over-differencing diagnostics from the lecture: the **variance** of the final series, and its **lag-1 autocorrelation**. Which route would you ship, and why? What does that tell you about the trend in this series?

PART D · FIND THE MISTAKES

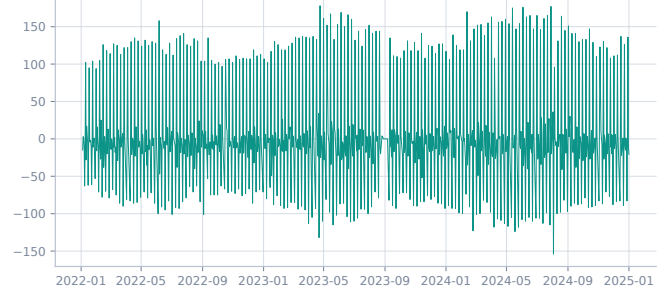
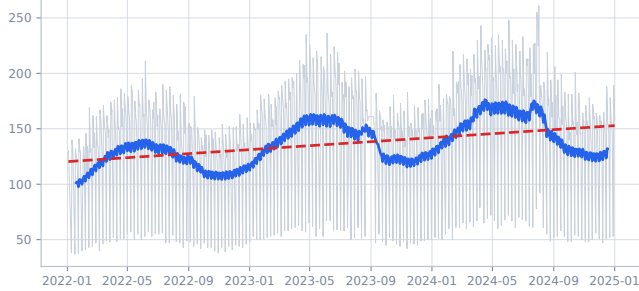
20 MIN · 12 MARKS

D1. Review this analysis

12 marks

A colleague sends you the following two-paragraph analysis of the clinic data, with these two figures. It contains **five** distinct errors. Find them, and for each one say what is wrong and what should have been done instead.

"The trend line rises 10.8 attendances per year, so we need 3 more staff by 2027. Differenced it once and ADF gives $p < 0.001$, so it is stationary"



THE COLLEAGUE'S WRITE-UP

"I fitted a straight line to the daily attendances. The slope is about 10.8 extra attendances per year and the fit is statistically significant with a very small p-value, so I am confident attendances will keep rising at that rate and we should budget for three more staff by 2027".

"I also checked stationarity. The raw series fails ADF, so I differenced it once. The differenced series gives ADF $p < 0.001$, which is well below 0.05, so it is stationary and ready for modelling. I did not need to transform anything because the mean is what matters. Finally I computed the mean and standard deviation of the whole series and flagged every day more than three standard deviations from the mean as an outlier. Not a single day was flagged, so the data is clean".

HINT

Two of the five are about the trend and the forecast. Two are about stationarity. One is about the outlier check. If you have a computer to hand, the second figure gives one of them away immediately: compute the ACF of that differenced series and look at lag 7.

STUCK?

Bring the question to the next lecture or post it on the forum. Questions that beat you are worth more class time than the ones that did not.